

Aairav Kalra

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Education

University of Waterloo

Honours Physics: Co-op, Computing Minor
Sept 2024 – Present

- Cumulative Average: 90.26%
- Term Distinction: 4x (All study terms)
- UW President's Scholarship

Skills

- **Python:** Used for coursework, personal projects and labs
- **Machine Learning and Data:** Training and evaluating ML models, scientific data analysis, lab reporting and data visualization. NumPy, Pandas, SciPy, XGBoost, Scikit-learn, Matplotlib.
- **Tools:** Linux (Course account management, scripting), Oracle Cloud VPS, Tailscale, Ollama, LM Studio, Openclaw, Claude Code, Codex, LaTeX, Racket, Vercel, Github/Git, Godot.
- **Teamwork and Collaboration:** Technical teaching and mentoring (office hours), coordinating TAs.

Projects and Activities

- **NASA Machine Learning Models:** Built and trained ML models (80+% accuracy) on NASA's Kepler and TESS data with web app integration to classify exoplanets as part of the NASA Space Apps Hackathon using SciKit-learn, XGBoost, Pandas and Numpy.
- **Boyajian's Star Flux Analysis (PHYS 267):** Analyzed FFI flux data for KIC 84628252 in Python using NumPy, SciPy and Matplotlib. Compared different fit models with the best piecewise fit (reduced $\chi^2 = 0.89$, $R^2 = 0.96$). Used regression, ANOVA and Kruskal-Wallis tests to show flux changes weren't a result of orientation.
- **PackVault:** Worked on the UI and UX of a collection tracking web app for trading cards with price integration aimed towards helping vendors manage and track inventory using Figma. Demo made using google antigravity. Created for vendors who usually have trouble tracking trades and inventory during card shows.
- **Remote Workstation and Local AI agent:** Set up a remote PC control system using Node.js Discord bot, Oracle Cloud Ubuntu VPS, Tailscale, Parsec, and a secured Flask Windows agent. Self-hosted a local LLM agent (Gemma via LM studio/Ollama) with a Telegram interface.
- **Humber Question Generator:** Generates questions based on unit analysis problems, uses adaptive difficulty to help students build their skillset. Built to help struggling students in Humber's Culinary Program.
- **Volunteering:** Earned Certificate of Appreciation (2x) for volunteering at the UW Physics Tutorial Centre for 200+ hours.
- **Leadership:** Led UW Quantum Club and North Park Astronomy Club as co-president. Organized events, coordinated with executives, collaborated with other clubs and IBM.
- **IBM Qiskit Fall Fest:** Collaborated with IBM to host the Fall 2026 Qiskit Fall Fest in Waterloo.

Experience

Instructional Support Assistant (Co-op), University of Waterloo

Sep 2026 – Dec 2026

- Created auto marking scripts and marking schemes for grading 600+ assignments, used Linux to maintain the course account. Served as the primary contact to students enrolled in CS 115, held office hours to assist with course material.
- Provided feedback to instructors regarding the exams and assignments, supervised lab sessions, assisted in proctoring and marking exams and assignments, coordinated tasks assigned to graduate teaching assistants.

Math Tutor (Co-op), Humber Polytechnic — North & Lakeshore Campuses

Jan 2026 – Apr 2026

- Delivered one-on-one, group, drop-in, and appointment-based tutoring in mathematics, calculus, statistics, and physics to college (Humber) and university (Guelph-Humber) students across 2 campuses.
- Received the highest rating amongst students (4.8/5 stars), earned CRLA certification, led review sessions, workshops, and exam prep for calculus and physics courses; created supplementary material using LaTeX.